

Exploiting Cross-QP Ordering in RDMA over Optical Fabrics

J. Meyer^(1,2), I. Tafur-Monroy⁽¹⁾, J. J. Vegas Olmos⁽²⁾

(1) Electrical Engineering Department, Eindhoven University of Technology, Eindhoven, The Netherlands

(2) Networking SW & Sys Arch, NVIDIA Corporation, Yokneam Illit, Israel

Abstract—Remote Direct Memory Access (RDMA) delivers microsecond-scale performance by bypassing the kernel, at the cost of the transport-layer reassembly that hides packet reordering from applications. The InfiniBand specification guarantees message ordering within a single Queue Pair (QP) and not across QPs; in practice, high-performance applications multiplex across many QPs and rely on cross-QP delivery remaining approximately FIFO in the common case. We quantify how much timing perturbation is needed to move that common case into one in which the application observes stale data, and we argue that the optical physical layer is the most plausible deployment site for such perturbation. Using a two-stage evaluation on real hardware and in software, we first establish that a production-class ConnectX-7 RoCEv2 pair already exhibits cross-QP corruption in a standard flag/data handoff (42 stale reads per 100,000 iterations at a 1,024-byte payload), then we mount a controlled delay-injection experiment on a software-RoCE (RXE) testbed and measure a dose-response curve. A single Gaussian-jittered `netem` rule on the sender's egress, whose mean delay we sweep and whose jitter scales with it, drives the corruption rate from 0.03 % at zero injected delay to 41.93 % at just 50 μ s of mean delay, approaching a 50 % asymptote past 200 μ s. The injector is a software analogue of the timing perturbation an optical-layer adversary can deploy in production (compromised ROADM, inline delay line, malicious OCS schedule). These measurements indicate that the cross-QP ordering gap is exploitable whenever an adversary has access to sub-millisecond timing control on one of several ECMP paths.

Index Terms—RDMA, RoCEv2, Optical Circuit Switching, Data Corruption

I. INTRODUCTION

Remote Direct Memory Access (RDMA) is now on the critical path of modern datacenter infrastructure: disaggregated memory, distributed storage, and AI training collectives all depend on it [1]. RDMA achieves microsecond-scale latency and line-rate throughput via kernel bypass, letting the Network Interface Card (NIC) read and write application memory directly. Eliminating the operating system's software transport layer also eliminates the safety net it provides, including the kernel reassembly buffer that allows TCP to hide reordering from applications.

Consequently, RDMA correctness depends on hardware ordering guarantees. The InfiniBand specification, which also governs RDMA over Converged Ethernet (RoCEv2), enforces strict in-order message delivery *within a single Queue Pair (QP)* using Packet Sequence Numbers (PSNs). The specification makes no ordering guarantees across different QPs.

In practice, high-performance RDMA applications multiplex operations across many QPs, because each QP maps to a different UDP source port and therefore, via Equal-Cost Multi-Path (ECMP) load balancing, to a different physical path. In benign environments approximate cross-QP FIFO ordering holds often enough to be usable as a common-case optimisation, and the resulting handoff patterns therefore rely on that common-case behaviour remaining close to FIFO. Our interest is in how much timing perturbation is needed to move these patterns out of the regime where that common-case holds.

This gap between specification and assumption is exploitable. An adversary with physical or control-plane access to the optical layer, for example via a compromised Reconfigurable Optical Add-Drop Multiplexer (ROADM), an inline fiber delay line, or a malicious Optical Circuit Switch (OCS) controller, can independently delay the wavelength carrying one QP's traffic while leaving its sibling untouched. Sub-microsecond precision is achievable, the perturbation leaves every packet intact, and there are no drops, so cryptographic integrity checks and loss-based monitoring never trigger.

Initial characterisations of this behaviour treat it as an ambient hazard: benign cross-QP skew already produces non-zero corruption on real hardware, and an adversary could in principle amplify that effect. In this paper we close the gap between “could in principle” and “does”. We run a controlled delay-injection experiment on a software-RoCE testbed and measure a dose-response curve from a benign 0.03 % baseline to 41.93 % at a 50 μ s mean injected delay, approaching a 50 % asymptote past 200 μ s. The injector is a direct software analogue of the optical-layer attack vector described above.

This paper makes three contributions:

- 1) A threat model for optical-layer adversarial timing manipulation against RDMA transports, covering vectors such as compromised ROADMs, inline delay lines, and malicious OCS controllers, together with their distinguishing properties of wavelength selectivity, invisibility to network-layer monitoring, and sub-microsecond precision.
- 2) An empirical demonstration on production-class ConnectX-7 RoCEv2 hardware that ambient cross-QP timing variability alone causes silent data corruption (42 per 100,000 iterations at 1,024-B payload) in a standard flag/data handoff; the single-QP control on the same hardware shows zero corruption. A causality-instrumented rerun confirms that 100 % of corrupted

iterations exhibit flag-before-data completion skew, while only 48.7 % of non-corrupted iterations do.

- 3) A controlled delay-injection experiment on a software-RoCE testbed that realises the optical attack in software. A single `tc netem` rule on the sender's egress drives the cross-QP corruption rate from 0.03 % at zero delay to 41.93 % at 50 μ s (a 1400-fold increase over the baseline), with a monotonic climb to a 50 % asymptote past 200 μ s. This is, to our knowledge, the first measured dose-response curve for RDMA ordering attacks.

II. BACKGROUND

A. RDMA Operations and Ordering

RDMA communication is built on the Queue Pair (QP) abstraction. Operations posted to a QP generate Completion Queue Entries (CQEs) reflecting the execution sequence within that context. Developers enforce intra-QP ordering using fence semantics (e.g., `IBV_SEND_FENCE`), which guarantee that all prior operations complete before a new one begins.

However, the InfiniBand specification limits *all* ordering guarantees to a single QP. There is no transport-level mechanism to synchronize or order messages sent across multiple QPs.

B. RoCEv2 and ECMP

RoCEv2 encapsulates RDMA traffic in UDP/IP. Unlike TCP, RoCEv2 handles out-of-order packets aggressively: if a packet arrives out of sequence, the receiver NIC discards it and triggers a go-back-N retransmission. To fully utilize datacenter bandwidth, applications use multiple QPs, each assigned a different UDP source port. Switches use these ports for ECMP routing, hashing different QPs to entirely separate physical paths. While Priority Flow Control (PFC) provides a lossless underlay, it does not prevent inter-path timing skew or reordering.

C. Optical Datacenter Networks

Modern datacenter topologies rely on optical components such as Optical Circuit Switches (OCS) and Reconfigurable Optical Add-Drop Multiplexers (ROADMs) to manage massive bandwidth. In these environments, different ECMP paths correspond to different wavelengths or optical circuits. OCS reconfigurations introduce variable latency, and propagation delays differ across wavelengths. An adversary with control-plane access to these components or physical access to the fiber can independently manipulate the delay of specific optical circuits, creating the conditions for precise cross-QP timing attacks.

III. RELATED WORK

A. RDMA Security

Recent scrutiny of RDMA has prioritized confidentiality and side-channel leakage over ordering integrity. ReDMARK [2] analyzed RDMA protocols, identifying flaws like predictable QP numbers and weak protection keys that enable packet injection and covert channels. Pythia [3] demonstrated remote

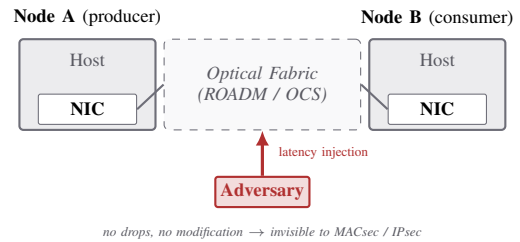


Fig. 1. Deployment topology and threat surface. Two RDMA nodes communicate through an optical fabric. An adversary with physical or control-plane access to that fabric injects latency on the wire, without modifying or dropping any packet. Fig. 2 develops the cross-QP attack mechanism.

side-channel attacks exploiting RNIC SRAM caching to deduce data access patterns. In contrast, our work explores a fundamental flaw in RDMA *integrity* and *ordering*: exploiting the gap between single-QP specification guarantees and multi-QP application assumptions to induce silent data corruption.

B. Network-Layer Reordering

Packet reordering and delay injection are established vectors at the transport and routing layers (e.g., TCP sequence manipulation [4], BGP hijacking [5]). TCP tolerates reordering by reassembling out-of-sequence segments in the kernel before delivering them to the application. RDMA has no equivalent reassembly step, so once a reordered packet is accepted by the NIC, it is visible to the application, and any cross-QP ordering the application depends on has to hold at the wire.

C. Optical-Layer Security

Optical security research historically focuses on passive eavesdropping (fiber tapping) and availability (in-band jamming) [6]. Our work extends this threat model to precise, wavelength-selective timing attacks. Because these microsecond-scale optical delays are invisible to higher-layer monitoring, they provide an ideal, stealthy mechanism to orchestrate the exact reordering conditions needed to break RDMA memory ordering.

IV. THREAT MODEL AND ATTACK SURFACE

This section defines the capabilities we assume of an optical-layer adversary and identifies the specific RDMA attack surface those capabilities expose.

A. Adversary Capabilities

We assume an adversary with physical or control-plane access to optical switching equipment in a datacenter or metropolitan area network. The adversary manipulates the physical transport of data without interacting with logical IP or Ethernet layers. Attack vectors include:

- **Optical Delay Lines:** Splicing a physical fiber spool into a link. A 100-meter splice introduces approximately 500 nanoseconds of fixed propagation delay.
- **Compromised ROADMs:** Reconfiguring a ROADM to dynamically route a specific wavelength through longer paths or extra amplification stages.

- **Malicious OCS Controllers:** Scheduling asymmetric, targeted circuit reconfigurations to introduce jitter during tear-down and setup phases.

These attacks inject sub-microsecond to millisecond delays with wavelength selectivity, meaning they target specific paths while leaving adjacent traffic unaffected. Because they operate purely in the time domain without modifying or dropping packets, they are completely invisible to IPsec, MACsec, and traditional network monitoring.

B. Attack Surface Decomposition

The core vulnerability is the **Cross-QP Ordering Gap**. The InfiniBand specification enforces in-order delivery within a single QP via PSNs, but specifies nothing about the relative ordering of messages across different QPs. High-performance applications such as KV stores and RPC frameworks use multiple QPs for parallelism and therefore rely on cross-QP delivery remaining close to FIFO in benign operation, even though nothing in the specification requires that property.

An adversary exploits this gap by leveraging ECMP routing. Because different QPs use distinct UDP source ports, ECMP hashing spreads them across different physical paths; in fabrics where those paths further map onto distinct wavelengths or optical circuits, an adversary at the optical layer can perturb one QP’s traffic in isolation. By selectively delaying the path carrying a data payload on QP_{data} while allowing a control flag on QP_{flag} to proceed, the adversary inverts the delivery order seen by the consumer. The application’s ordering expectation no longer holds, and the result is silent data corruption or race conditions in the subsequent memory access.

Secondary vulnerabilities include:

- **Intra-QP Denial-of-Service:** Adversarial delay within a single QP forces continuous go-back-N retransmissions, crashing throughput and potentially forcing the QP into an irrecoverable error state.
- **Fence Semantics:** Delaying packets artificially widens timing windows during lock acquisition, inducing distributed race conditions that bypass atomic primitives.

Unlike TCP, which buffers out-of-sequence packets in the kernel and reassembles them before delivering to the application, RDMA writes payload data directly to user-space memory. Once a packet passes NIC hardware validation, its data is visible to the application immediately, without any software-level reordering check.

V. EXPERIMENTAL DEMONSTRATION

We first confirm on production ConnectX-7 (CX7) RoCEv2 hardware that benign cross-QP timing variability alone produces silent data corruption, then we run a controlled delay-injection sweep on a software-RoCE testbed and show that an adversary with sub-millisecond timing control drives the corruption rate from parts-per-million to near 50 %.

A. Testbed Setup

We use two back-to-back pairs of VMs. Testbed A establishes realism: NVIDIA ConnectX-7 SmartNICs running

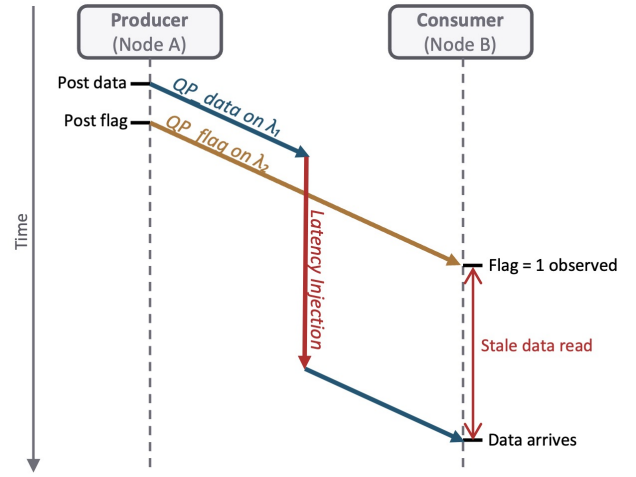


Fig. 2. Cross-QP attack sequence. The producer posts an RDMA WRITE of the data payload on QP_{data} (carried by λ₁) followed by a 1-byte flag on QP_{flag} (carried by λ₂). Adversarial latency injection on λ₁ delays the data packet so the flag arrives first; the consumer reads the memory region during the resulting stale-data window, before the real data has landed.

```
// Producer side: Two separate QPs
ibv_post_send(qp_data, &wr_data, &bad_wr);
ibv_post_send(qp_flag, &wr_flag, &bad_wr);

// Consumer side: The vulnerable polling loop
while (__atomic_load_n(flag_buf, __ATOMIC_ACQUIRE) == 0) {
    sched_yield(); // Spin waiting for flag
}
// VULNERABILITY: Consumer assumes data_buf is valid!
process_payload(data_buf);
```

Fig. 3. Vulnerable cross-QP handoff. The consumer assumes the data payload is complete when the flag arrives, which is unsafe when the operations are carried by different QPs.

RoCEv2 on a production-class NIC offload path. Testbed B enables controlled delay injection: the Linux rdma_rxe soft-RoCE transport [7], which implements RoCEv2 entirely in the kernel and therefore lets us shape egress traffic with Linux netem [8]. On both testbeds we use custom libibverbs programs calling ibv_post_send and ibv_poll_cq directly, so transport-level behaviour is not obscured by middleware.

B. Hardware Realism: Ambient Cross-QP Corruption

On Testbed A, we execute the flag/data handoff pattern used by high-performance RDMA services such as HERD [9], Pilaf [10], and FaRM [11]. The producer issues an RDMA WRITE of the data payload on QP_{data}, immediately followed by an RDMA WRITE of a 1-byte flag on QP_{flag}. The consumer spins on the flag and, upon its state change, reads the data payload; it reports corruption if the value read is stale. This vulnerable cross-QP handoff is illustrated in Fig. 3.

We run this workload for 20,000 iterations per run, five runs per configuration, across payload sizes of 64 B, 1,024 B, and 64 KB. No netem rule is installed on either endpoint. Ambient cross-QP skew alone is sufficient to drive corruption

TABLE I
AMBIENT STALE-READ CORRUPTION ON CX7 (TESTBED A). NO
INJECTED DELAY; 100,000 ITERATIONS TOTAL PER PAYLOAD.

Payload Size	Cross-QP Corruption	Single-QP Control
64 B	0 / 100,000 (0.000%)	0 / 100,000 (0.000%)
1,024 B	42 / 100,000 (0.042%)	0 / 100,000 (0.000%)
65,536 B	1 / 100,000 (0.001%)	0 / 100,000 (0.000%)

at the 1,024 B payload; the single-QP control (both operations on one QP) sees zero corruption, consistent with the spec’s intra-QP PSN guarantee (Table I).

To link these events mechanistically to cross-QP timing inversion, we run a 300,000-iteration causality campaign at the 1,024-B payload, recording per-QP completion timestamps. We observe 28 stale-read events (0.0093 %); the lower rate compared with Table I is consistent with run-to-run variability in ambient timing skew, which is sensitive to background load and kernel scheduling jitter. Every corrupted iteration occurred when the flag-QP completion preceded the data-QP completion ($t_{\text{flag}} < t_{\text{data}}$), whereas only 48.7 % of non-corrupted iterations showed that ordering. The 100 % conditional alignment points to flag-before-data skew as the cause of the observed corruption. Timestamps here come from host-side software polling because the CQE hardware-timestamp facility was unavailable in this environment, so we treat the causality result as directional rather than quantitative.

C. Controlled Delay-Injection Attack

We reproduce and control the ambient effect on Testbed B. Because `rdma_rxe` is a pure kernel-path transport, we can apply `tc netem` to the sender’s data-plane NIC and verify interception on the `qdisc` counter. For each delay point X in $D = \{0, 50, 100, 150, 200, 300, 500, 1000\}$ μs , we install a `netem` rule of the form `delay X 0.4X distribution normal limit 10000` on the sender’s egress and run the same flag/data handoff workload for 10,000 iterations at 1,024 B payload, repeating 5 times per delay point for a total of 400,000 trials. The injected delay has a Gaussian jitter component of $0.4X$ specifically because fixed delay alone preserves packet ordering, whereas real optical delay mechanisms (wavelength-dependent propagation, ROADM re-configuration, OCS switchover jitter) introduce variance. Each trial’s consumer-side corruption count is recorded from the receiver; we verify `netem` interception by checking that the `qdisc` reports tens of thousands of intercepted packets per run.

Figure 4 shows the resulting delay-to-corruption curve. Three observations anchor the argument. First, the *baseline* ($X = 0$, no `netem`) produces 0.030 % corruption (15 in 50,000, 95 % CI [0.017, 0.049] %), within a factor of 1.4 of the CX7 ambient rate at the same payload, confirming that the software testbed reproduces the hardware behaviour when the injector is off. Second, at just 50 μs mean injected delay the rate jumps to **41.93 %** (95 % CI [41.50, 42.36] %), a 1400-fold increase over the baseline. Third, the curve approaches a 50 % asymptote as delay grows past 200 μs , reaching 49.67 %

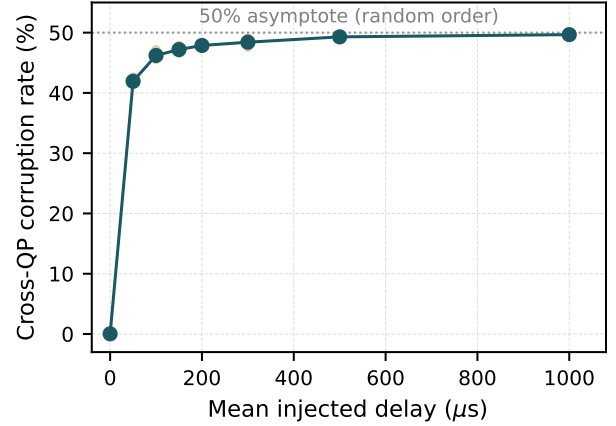


Fig. 4. Cross-QP corruption rate on Testbed B (RXE) vs. mean injected `netem` delay (Gaussian jitter = $0.4 \times$ mean). Small markers are individual reps (5 per delay, 10,000 iterations each); large markers are the pooled mean with 95 % exact binomial confidence intervals. The zero-delay baseline (0.030 %) matches the CX7 ambient rate (0.042 %) at the same payload. A sharp threshold at 50 μs drives the rate to 41.93 % (a 1400-fold increase over baseline); the curve then approaches a 50 % asymptote, the value expected when the injected jitter is so large that the data and flag packets arrive in random order.

at $X = 1$ ms. This asymptote is the rate we would expect if the flag and data packets arrived in independent random order: at that point each iteration is a fair coin flip, and the adversary wins the race in approximately half of all trials.

D. Interpretation

These results demonstrate, end-to-end, the threat of Section IV. A small, Gaussian, sub-millisecond timing perturbation on the sender’s egress is sufficient to convert a benign multi-QP handoff into a silent data-corruption channel at a severe rate: roughly half of all handoffs return stale data. The single-QP control on the same testbed continues to show zero corruption, confirming that the failure mode is exactly the cross-QP ordering gap identified in Section IV, not a transport-level bug.

The injector we use in Testbed B (`tc netem delay X 0.4X distribution normal`) is a software analogue of the exact mechanism an optical-layer adversary would deploy: a variable sub-millisecond delay applied selectively to traffic on a particular path. Section VI argues that the optical layer is both the realistic deployment site for this mechanism and the one at which it is hardest to detect. The controlled sweep here shows that once such an injector is in place, the cross-QP ordering gap is not a latent hazard but an actively exploitable attack surface.

VI. OPTICAL ATTACK FEASIBILITY

The controlled sweep of Section V-C demonstrates the attack in software. In production, the realistic deployment site for exactly this timing perturbation is the optical physical layer, where delay can be applied with sub-microsecond precision, wavelength selectivity, and no loss signature.

A. Mapping the Software Injector to Optical Mechanisms

The `tc netem delay X 0.4X` distribution normal rule used in Section V-C is a software stand-in for three concrete optical-layer mechanisms, each of which produces a qualitatively equivalent perturbation on one ECMP path while leaving others untouched:

- **Inline fiber delay line.** Splicing a fiber spool into the target path introduces fixed propagation delay: single-mode fiber carries light at roughly $5 \mu\text{s}/\text{km}$, so 100 m of fiber adds about 500 ns. A few hundred metres of coil produces the same 100–1000 μs mean delays that our sweep shows drive the corruption rate to near 50 %. The fiber coil introduces essentially zero additional jitter; a stealthier adversary combines a fiber delay with a low-amplitude, variable-gain element to produce the Gaussian-jittered perturbation we emulate in `netem`.
- **Compromised ROADM.** A ROADM under adversarial control can route a specific wavelength through an extra amplification stage or a physically longer intra-node path, and can modulate that routing dynamically. Where ECMP paths are carried on distinct wavelengths, this provides both selective targeting of a single ECMP path, and therefore in practice a single QP, and the time-varying perturbation that our sweep shows is sufficient.
- **Malicious OCS controller.** In OCS-based datacentre architectures (e.g. Jupiter [12], Sirius [13]), circuit reconfiguration scheduling is under software control. An adversary with control-plane access schedules asymmetric reconfigurations on the circuits carrying the target application’s QP_{data} , introducing microsecond- to millisecond-scale jitter during tear-down and setup, in the same jitter regime that drives our measured curve to its asymptote.

Empirically, the corruption rate rises from 0.03 % to 41.9 % over a 50 μs mean delay step in our sweep. All three optical mechanisms above can achieve that step trivially; the question is not feasibility but concealment.

B. The Difficulty of Detection

Optical-layer timing attacks are nearly invisible to conventional security infrastructure. Cryptographic integrity checks (MACsec, payload CRCs, IPsec ESP) provide no defence: the packets arrive mathematically intact and properly authenticated. Loss-based anomaly detectors do not trigger because nothing is dropped. Standard network telemetry, which is per-packet or per-flow and coarse-grained, has no baseline against which to flag a 50 μs perturbation on one of several ECMP paths, a delta indistinguishable from ordinary queuing or link-level jitter.

Distinguishing a malicious optical perturbation from benign variation requires cross-flow latency correlation at the nanosecond level, instrumentation entirely absent from standard RDMA deployments and fundamentally in tension with the kernel-bypass design that motivates RDMA in the first place.

VII. MITIGATIONS

Because the measured attack converts a 0.03 % background rate into a ~ 50 % corruption rate with sub-millisecond delay, mitigations that merely improve ambient behaviour are insufficient. Defences must assume an active adversary with sub-microsecond timing control on at least one ECMP path.

A. Application-Level Defences

The most direct mitigation is to stop depending on cross-QP order at the application layer. Cross-QP protocols should carry explicit end-to-end sequence numbers and validate them in software before consuming the handoff data. Equivalently, a unified cryptographic checksum computed over the data payload *and* its control flag rejects any stale-data/current-flag combination at consume time.

The most robust choice — and the one the single-QP control in Section V-B shows is safe — is to keep ordering-critical handoffs on a single QP, trading parallelism for hardware-enforced PSN ordering.

B. Transport-Level Defences

Future NICs should expose cross-QP synchronisation primitives by extending fence semantics so that a fence on QP_{flag} can wait for prior operations on QP_{data} . NIC firmware could also maintain inter-arrival jitter histograms per ECMP path and raise alerts when one path’s tail diverges from its siblings, which is precisely the signature our 50 μs perturbation produces. Because our measured attack relies on live packet delay rather than replay, encrypted transports such as IPsec are only effective with strict per-packet freshness bounds; loose freshness windows do not help.

C. Physical-Layer Defences

Securing the optical underlay is essential given the evidence in Section V-C that a single sub-millisecond delay perturbation is sufficient. Continuous Optical Time Domain Reflectometry (OTDR) detects inline splices such as fiber delay lines. Path-delay verification protocols baseline each wavelength’s propagation delay and flag unexplained shifts. Finally, operators must enforce strict control-plane security for ROADMs and OCS controllers. Our attack works at whatever delay resolution the fabric can produce, so hardening the fabric’s control plane is the first line of defence against the most flexible adversary.

VIII. CONCLUSION

The InfiniBand specification enforces ordering only within a single QP; cross-QP ordering is a statistical property of benign networks rather than a guarantee. We have quantified how much that property depends on benign timing. On production ConnectX-7 RoCEv2 hardware, ambient cross-QP timing skew already produces measurable silent data corruption in a standard flag/data handoff. On a software-RoCE testbed, a single Gaussian-jittered `netem` rule on the sender’s egress, a software stand-in for the timing perturbation an optical adversary deploys in production, drives the corruption rate

from 0.03 % at zero injected delay to 41.9 % at just 50 μ s, approaching a 50 % asymptote past 200 μ s. Because this perturbation is wavelength-selective, loss-free, and invisible to cryptographic integrity checks, the optical physical layer is a plausible deployment site for it and a difficult one at which to detect it. Addressing this vulnerability requires explicit recognition of the cross-QP ordering gap and coordinated defences at the application, transport, and optical layers; the measurements reported here are intended to inform the priorities of each.

ACKNOWLEDGMENT

This work was funded by the QUARC project of the European Union (EU) Horizon Europe research and innovation programme within the framework of Marie Skłodowska-Curie Actions under grant number 101073355. Generative AI tools were used to assist with literature review, drafting, and editing of this manuscript; all technical content was verified by the authors.

REFERENCES

- [1] Z. Hu *et al.*, “Demystifying nccl: An in-depth analysis of gpu communication protocols and algorithms,” 2025.
- [2] B. Rothenberger, K. Taranov, A. Perrig, and T. Hoefler, “ReDMark: Bypassing RDMA security mechanisms,” in *30th USENIX Security Symposium (USENIX Security 21)*. USENIX Association, Aug. 2021, pp. 4277–4292.
- [3] S.-Y. Tsai, M. Payer, and Y. Zhang, “Pythia: Remote oracles for the masses,” in *28th USENIX Security Symposium (USENIX Security 19)*. Santa Clara, CA: USENIX Association, Aug. 2019, pp. 693–710.
- [4] S. M. Bellovin, “Security problems in the TCP/IP protocol suite,” *Computer Communication Review*, vol. 19, no. 2, pp. 32–48, Apr. 1989.
- [5] P. Sermpetis, V. Kotronis, A. Dainotti, and X. Dimitropoulos, “A survey among network operators on bgp prefix hijacking,” *Computer Communication Review*, vol. 48, no. 1, 2018.
- [6] V. Spurny, P. Munster, A. Tomarov, T. Horvath, and E. Skaljko, “Physical layer components security risks in optical fiber infrastructures,” *Sensors*, vol. 22, no. 2, p. 588, 2022.
- [7] Linux Kernel Project, “Soft RoCE driver (drivers/infiniband/sw/rxe),” <https://www.kernel.org/doc/html/latest/infiniband/index.html>, accessed 2026-04-17.
- [8] S. Hemminger, “Network emulation with NetEm,” in *Proceedings of the 6th Australia’s National Linux Conference (linux.conf.au)*, Canberra, Australia, Apr. 2005.
- [9] A. Kalia, M. Kaminsky, and D. G. Andersen, “Using rdma efficiently for key-value services,” in *Proceedings of the 2014 ACM Conference on SIGCOMM*. Chicago, Illinois: ACM, 2014, pp. 295–306.
- [10] C. Mitchell, Y. Geng, and J. Li, “Using One-Sided RDMA reads to build a fast, CPU-Efficient Key-Value store,” in *2013 USENIX Annual Technical Conference (USENIX ATC 13)*. San Jose, CA: USENIX Association, Jun. 2013, pp. 103–114.
- [11] A. Dragojević, D. Narayanan, M. Castro, and O. Hodson, “FaRM: Fast remote memory,” in *11th USENIX Symposium on Networked Systems Design and Implementation (NSDI 14)*, Seattle, WA, 2014, pp. 401–414.
- [12] L. Poutievski *et al.*, “Jupiter evolving: Transforming google’s datacenter network via optical circuit switches and software-defined networking,” in *Proceedings of the ACM SIGCOMM 2022 Conference*. Amsterdam, Netherlands: ACM, 2022.
- [13] H. Ballani *et al.*, “Sirius: A flat datacenter network with nanosecond optical switching,” in *Proceedings of the ACM SIGCOMM Conference on Data Communication*, Aug. 2020.